

AI-Powered Task & Issue Manager (Smart Jira)

1. Introduction

The AI-Powered Task & Issue Manager is a web-based project management application developed using the MERN Stack (MongoDB, Express.js, React.js, Node.js). The system helps teams manage projects, tasks, and team collaboration efficiently. It also incorporates Artificial Intelligence to automatically generate and distribute tasks among team members based on their roles.

2. Problem Statement

Traditional task management systems require managers to manually create and assign tasks, which can be time-consuming and error prone.

The proposed system automates task generation and assignment using AI while still providing manual control to users.

3. Objectives

- Manage projects efficiently
- Create and track tasks
- Assign tasks to team members
- Support role-based collaboration
- Generate tasks automatically using AI
- Visualize tasks through Kanban Board
- Monitor project progress using Dashboard

4. Technologies Used

Frontend

- React.js
- Tailwind CSS
- Axios
- React Router DOM

Backend

- Node.js
- Express.js

Database

- MongoDB Atlas
- Mongoose

Authentication

- JWT Authentication
- bcrypt.js

AI Integration

- Google Gemini API

5. System Features

Authentication

- User Registration
- User Login
- JWT Authentication

Project Management

- Create Project
- Update Project
- Delete Project
- View Projects

Team Collaboration

- Add Members
- Role-Based Users
- Member Management

Task Management

- Create Task
- Update Task Status
- Delete Task
- Task Assignment

Dashboard

- Total Projects
- Total Tasks
- Todo Count
- In Progress Count
- Completed Count

AI Features

- AI Task Generation
- AI Task Distribution
- Role-Based Assignment

6. User Roles

Role	Responsibility
Frontend Engineer	UI Development
Backend Engineer	API Development
QA Engineer	Testing
Full Stack Engineer	Frontend + Backend
DevOps Engineer	Deployment

7. Database Design

User Collection

```
{
  name,
  email,
  password,
  role
}
```

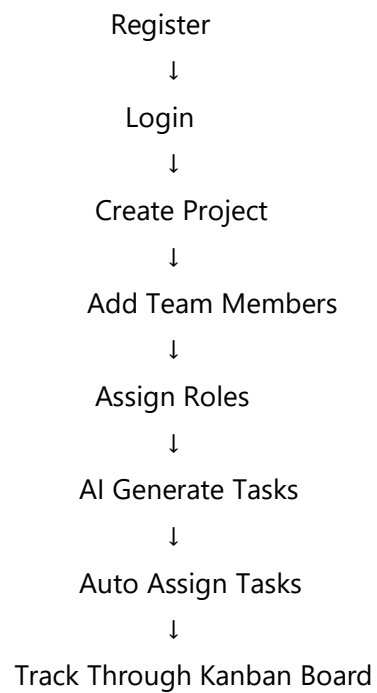
Project Collection

```
{
  title,
  description,
  owner,
  members
}
```

Task Collection

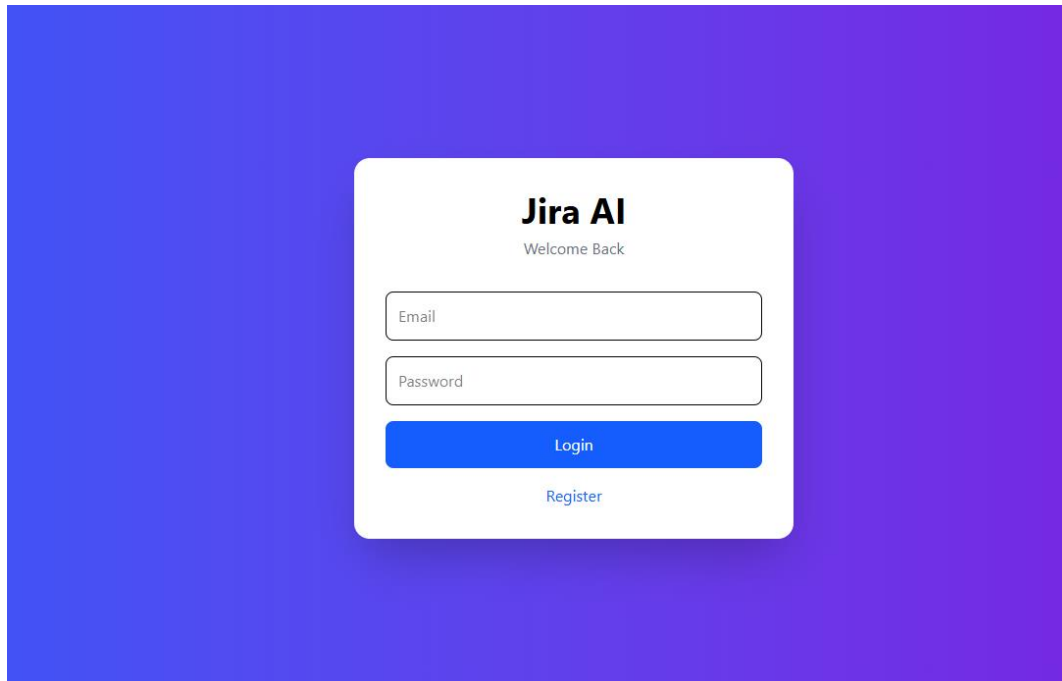
```
{  
  
title,  
description,  
status,  
priority,  
project,  
assigned To  
}
```

8. System Workflow



9. Screenshots

Login Page



The login page features a white card centered on a blue-to-purple gradient background. The card contains the 'Jira AI' logo, a 'Welcome Back' message, and input fields for 'Email' and 'Password'. A blue 'Login' button is positioned below the password field, and a 'Register' link is at the bottom of the card.

Jira AI
Welcome Back

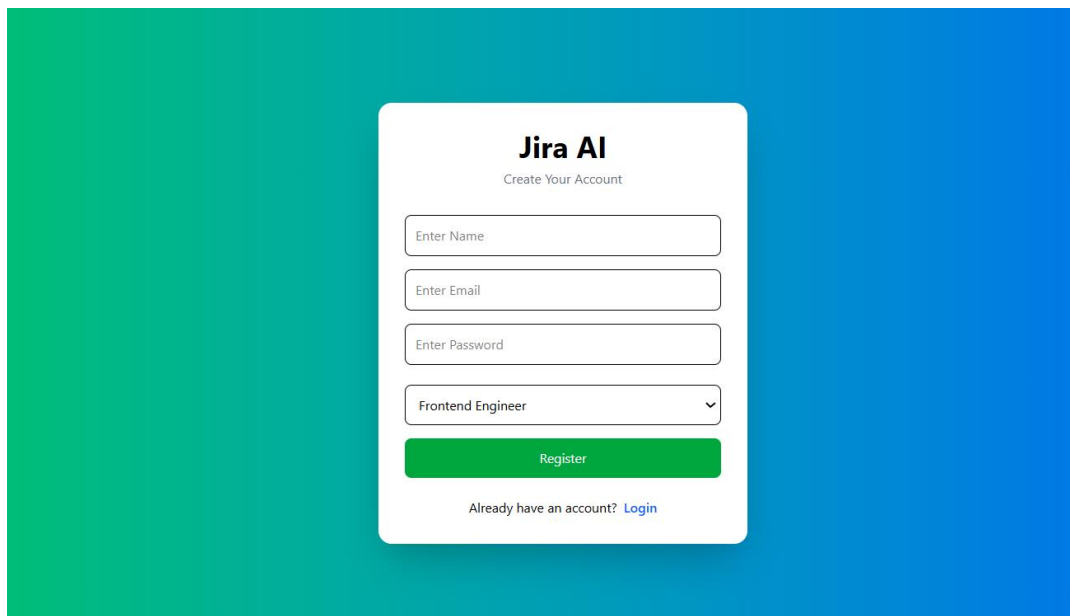
Email

Password

Login

[Register](#)

Register Page



The register page features a white card centered on a green-to-blue gradient background. The card contains the 'Jira AI' logo, a 'Create Your Account' message, and input fields for 'Enter Name', 'Enter Email', and 'Enter Password'. A dropdown menu for 'Frontend Engineer' is located below the password field. A green 'Register' button is positioned below the dropdown, and a link to 'Login' is at the bottom of the card.

Jira AI
Create Your Account

Enter Name

Enter Email

Enter Password

Frontend Engineer ▼

Register

Already have an account? [Login](#)

Dashboard

Jira AI

Dashboard

Projects

Logout

Search...

Create Project

Project Title

Project Description

Create Project

Projects (4)

spotify Clone

Final Year Project

Members

AI Generate Tasks

Member Email

Add Member

OpenDelete

AI Jira Clone

Final Year Project

Members

AI Generate Tasks

Member Email

Add Member

OpenDelete

backend

Create Backend

Members

AI Generate Tasks

Member Email

Add Member

OpenDelete

Tasks Board

Project Task Board

Manage project tasks efficiently

Create Task

Task Title

Task Description

Medium

Select Member

Create Task

Todo (1)

Login

Login page

Assigned To: Gaurav

High

Todo

Delete

In Progress (1)

Register

Register Page

Assigned To: Gaurav

Medium

In Progress

Delete

Done (1)

Password

Password Update

Assigned To: Gaurav

Low

Done

Delete

10. Future Scope

- Email Notifications

- Gantt Charts
- Team Performance Analytics
- AI Progress Prediction
- AI Risk Analysis

11. Conclusion

The AI-Powered Task & Issue Manager successfully combines project management, team collaboration, and artificial intelligence to improve productivity. The system automates task generation and assignment while maintaining flexibility through manual task management options.